VINAY M MADGI

Computer Science Engineering Student

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.E	KLE Technological University, Hubballi	8.36	2023–2027
Class 12	K.H Patel PU Science College, Hubballi	70%	2022
Class 10	K.L.E Society's English Medium School, Hubballi	68%	2020

PROJECTS

AI Academic Report Generation System
Tech Stack: Python, PyTorch, LangChain, Streamlit, FAISS, Transformers

- Built an end-to-end academic report generation pipeline using LLMs, RAG, and FAISS-based retrieval for citation-grounded content synthesis.
- Designed a hierarchical planning + reflection workflow that automatically regenerates low-confidence sections using evidence support and citation coverage checks.
- Integrated NLI-based hallucination detection using transformer models like DeBERTa-MNLI and SciFact evaluation for factual consistency validation.
- Developed a responsive Streamlit web application with report history management, document export, and automated citation handling.

Meta-Learning Based Few-Shot Classification
Tech Stack: Python, PyTorch, ProtoNet

- Implemented a few-shot meta-learning framework using Prototypical Networks (ProtoNet) with PyTorch for N-way K-shot classification tasks.
- Built a custom meta-task generator for dynamic episodic training on Omniglot and CIFAR-100 datasets.
- Designed and trained a deep CNN embedding architecture with static hyperparameter optimization across multiple meta-learning tasks.
- Performed comparative evaluation of optimizers and configurations using automated experiment tracking, result visualization, and performance analysis.

Sports Inventory Management System
Tech Stack: Express.js, Node.js, React.js, MongoDB (MERN Stack)

- Developed a full-stack MERN application to manage and track sports inventory in real-time.
- Implemented RESTful APIs using Node.js and Express.js for efficient backend operations.
- Designed responsive UI using React for seamless user interaction and data visualization.
- Integrated MongoDB for scalable data storage and CRUD operations.

SKILLS

Languages: C++, Java, Python, SQL
Web Technologies: Express.js, HTML, MongoDB
Core Concepts: Data Structures and Algorithms, OOPs, DBMS, Operating Systems
Tools: Git, GitHub, Websockets

ACHIEVEMENTS

Research Paper Presentation: Presented a paper titled "Efficient Image-Text Alignment for Fashion Data Using Frozen Vision-Language Encoders" at the 5th International Conference on Advanced Computing and Intelligent Technologies (ICACIT-2026), Netaji Subhas University of Technology (NSUT), New Delhi.